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Unit “ 4 : Session “ 8 : SLO “ 2 | Micro-programmed Control “ Choose the Correct Answer

■ I. Multiple Choice

1. b) Control Store (ROM) “ microinstructions reside in a dedicated control store inside the CPU.
2. b) A single control signal “ in horizontal microcode each bit drives one control line directly.
3. b) Micro-PC “ the micro program counter holds the address of the next microinstruction.
4. b) Shorter and encoded “ vertical microcode packs signals into encoded fields, so words are narrow but need decoding.
5. c) Determine next microinstruction address “ the microsequencer handles microprogram flow (sequencing/branching).
6. b) Writable Control Store (WCS) “ a writable (RAM) control store allows changing microprograms at run time (dynamic microprogramming).
7. a) Smaller control steps within microinstructions “ nanocode is a second level below microcode, further breaking microinstructions into nano-steps.
8. b) Complex CISC instructions “ microprogramming makes complicated multi-step instructions manageable.
9. b) Easier modification “ changing the control unit means editing microcode rather than rewiring hardware.
10. b) Emulation “ using microcode to simulate another (complex) instruction set is called emulation.